

CMSC828G: Special Topics in Systems for Machine Learning

Parallel Training

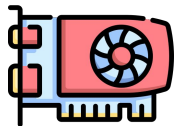
Instructor: Yafan Huang



DEPARTMENT OF
COMPUTER SCIENCE

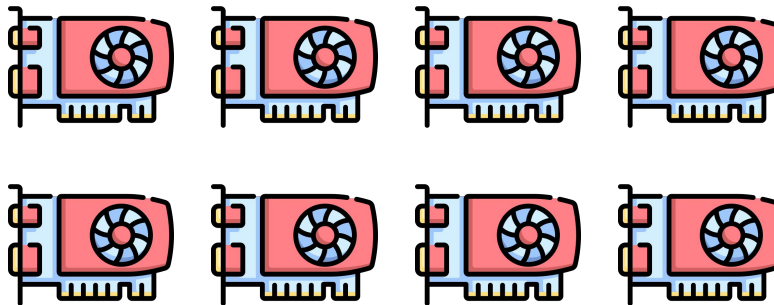
Why Parallel Training?

- LLM training requires substantial GPU memory and computational resources.
- Parallel training distributes workloads across multiple GPUs in a diverse way to enable larger models and faster training.



Single GPU

Out of memory



Multiple GPUs

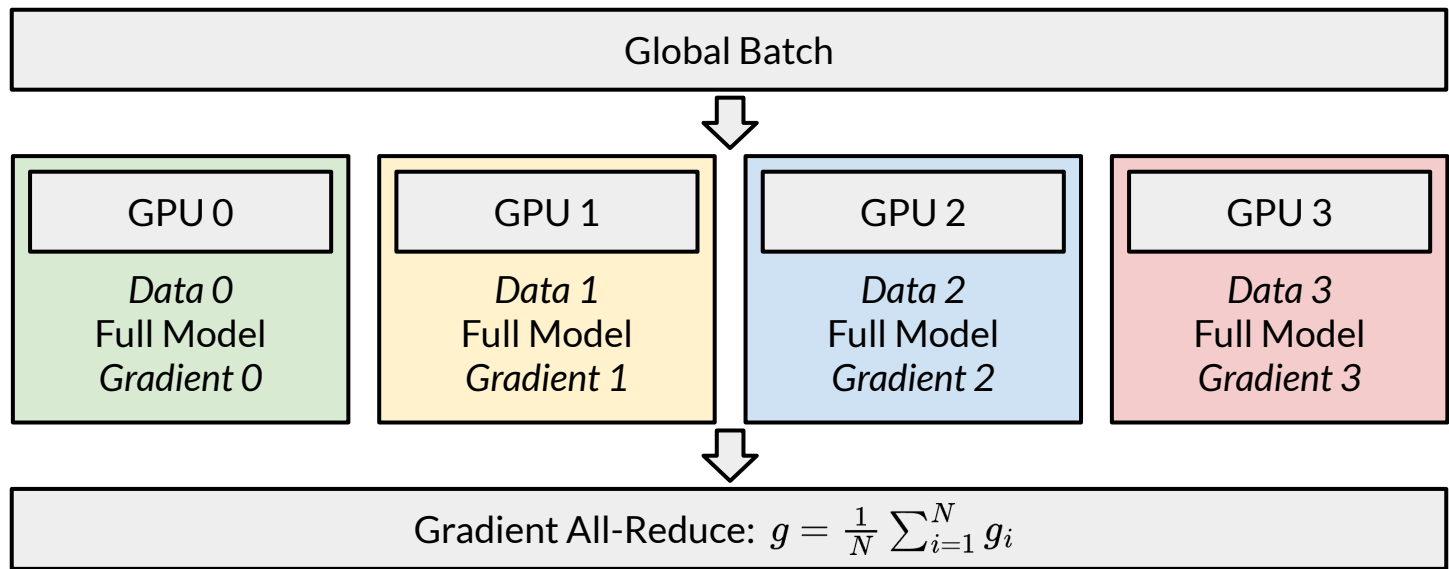
Distributed Training

Dimensions of Parallelism

- Parallel training partitions the workload across different dimensions.
- Different parallelism strategies address different computational and memory bottlenecks.
 - Data Parallelism (DP) - batch dimensions
 - Tensor Parallelism (TP) - tensor dimensions
 - Pipeline Parallelism (PP) - model depth
 - Context parallelism (CP) - sequence length
 - Expert parallelism (EP) - MoE experts

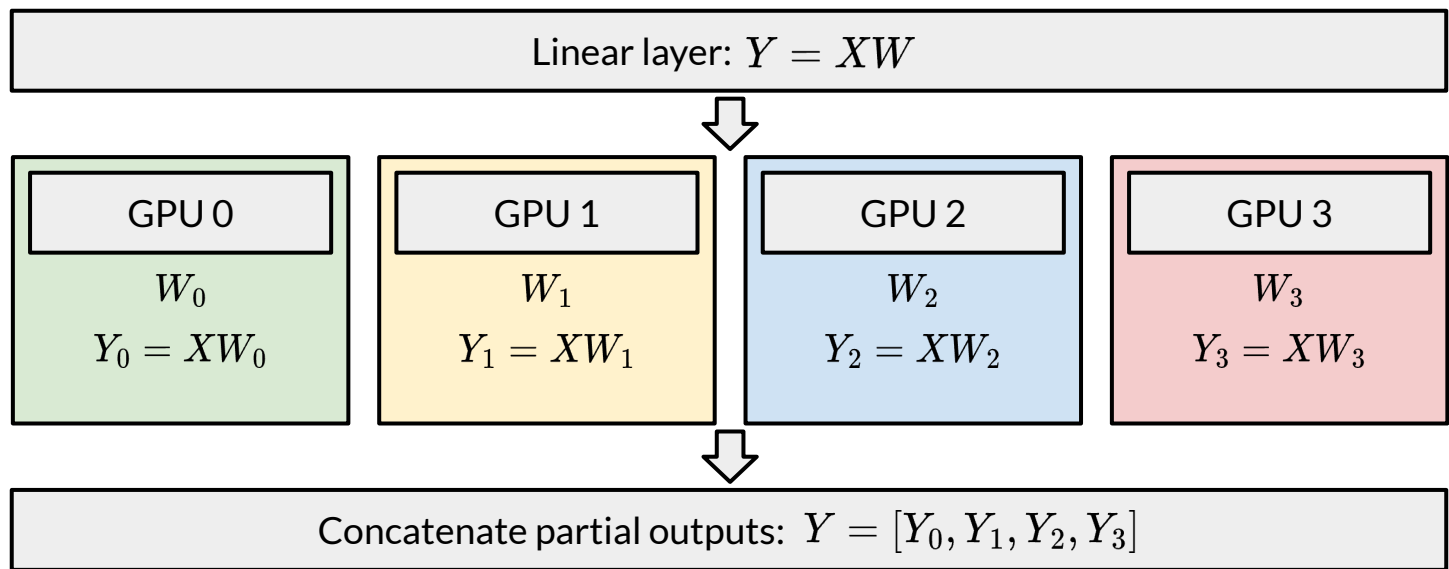
Data Parallelism (DP)

- Each GPU maintains a full model replica and processes a different subset of training data, and gradients are synchronized across GPUs using *All-Reduce* before updating model parameters.



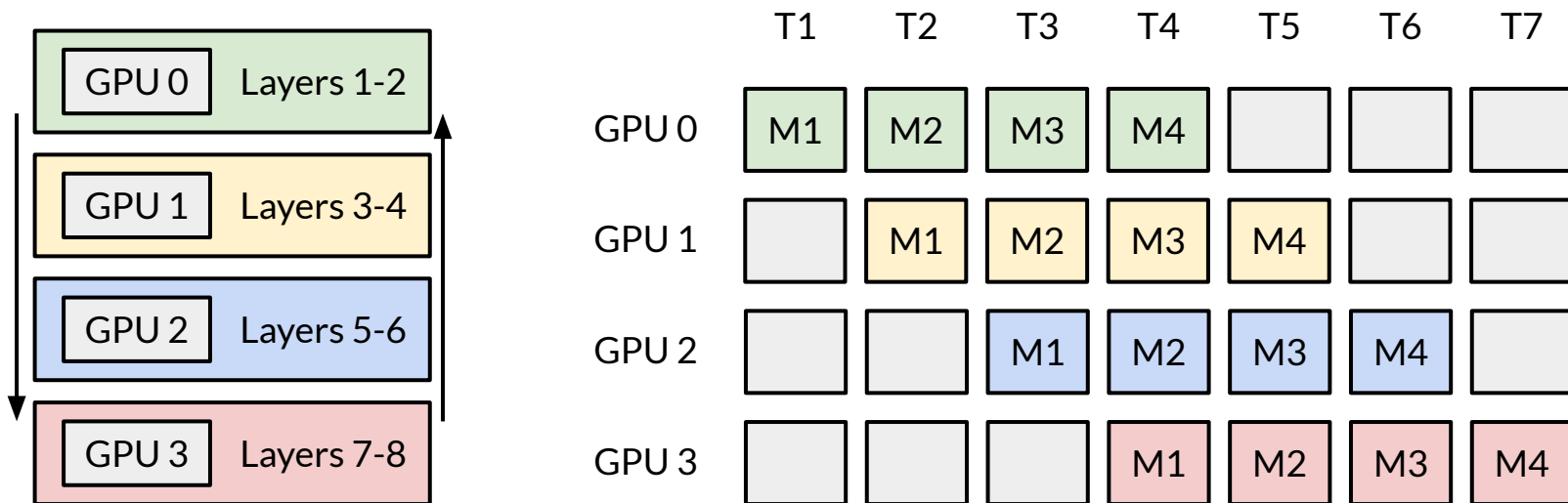
Tensor Parallelism (TP)

- Tensor parallelism partitions the parameters and computation within an individual model layer across multiple GPUs, and GPUs compute partial outputs and communicate to produce the required layer output.



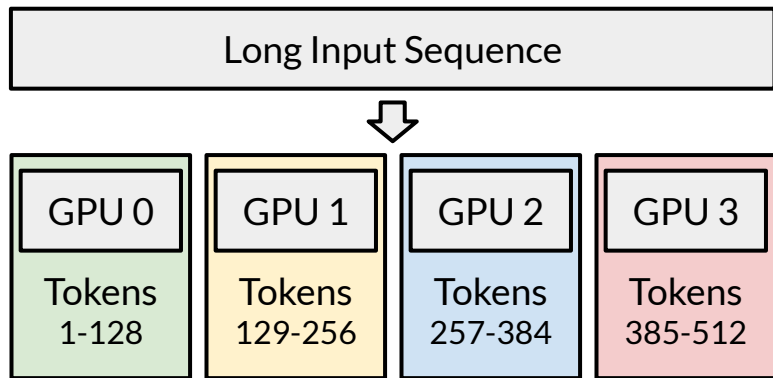
Pipeline Parallelism (PP)

- Pipeline parallelism partitions a model into sequential stages, each assigned to a different GPU. Activations and gradients are communicated between neighboring stages during forward and backward passes.

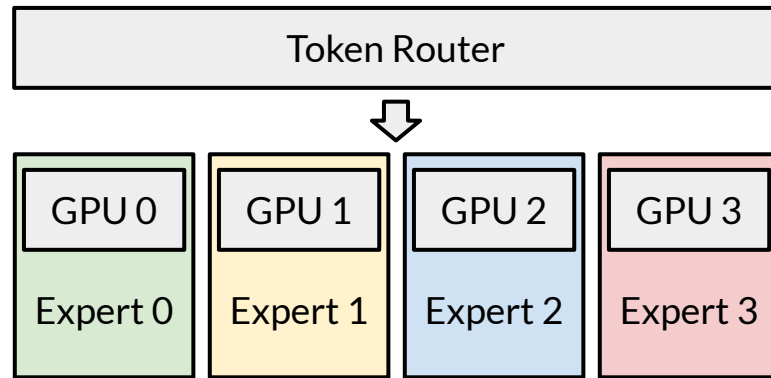


Context Parallelism and Expert Parallelism

- Context Parallelism (CP) partitions long sequences across GPUs to distribute computation and activation memory.
- Expert parallelism (EP) distributes different experts in a Mixture-of-Experts (MoE) model across GPUs.



CP: Cross-GPU Attention Communication



EP: Token Dispatch and Combine

Student Presentations

- [VLDB'23] PyTorch FSDP: Experiences on Scaling Fully Sharded Data Parallel
- [OSDI'25] WLB-LLM: Workload-Balanced 4D Parallelism for Large Language Model Training